Lecture 20: Max-Flow II

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601.433/633 Introduction to Algorithms

Introduction

Last time:

- Max-Flow = Min-Cut
- Can compute max flow and min cut using Ford-Fulkerson: while residual graph has an $s \to t$ path, push flow along it.
  - Corollary: if all capacities integers, max-flow is integral
  - If max-flow has value F, time $O(F(m+n))$ (if all capacities integers)
  - Exponential time!

Today:

- Important setting where FF is enough: max bipartite matching
- Two ways of making FF faster

Max Bipartite Matching

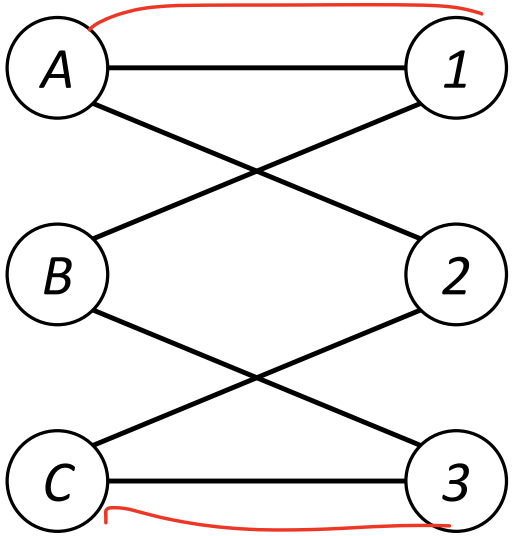
Setup

Definition

A graph $G = (V, E)$ is *bipartite* if V can be partitioned into two parts L, R such that every edge in E has one endpoint in L and one endpoint in R.

Definition

A *matching* is a subset $M \subseteq E$ such that $e \cap e' = \varnothing$ for all $e, e' \in M$ with $e \neq e'$ (no two edges share an endpoint)



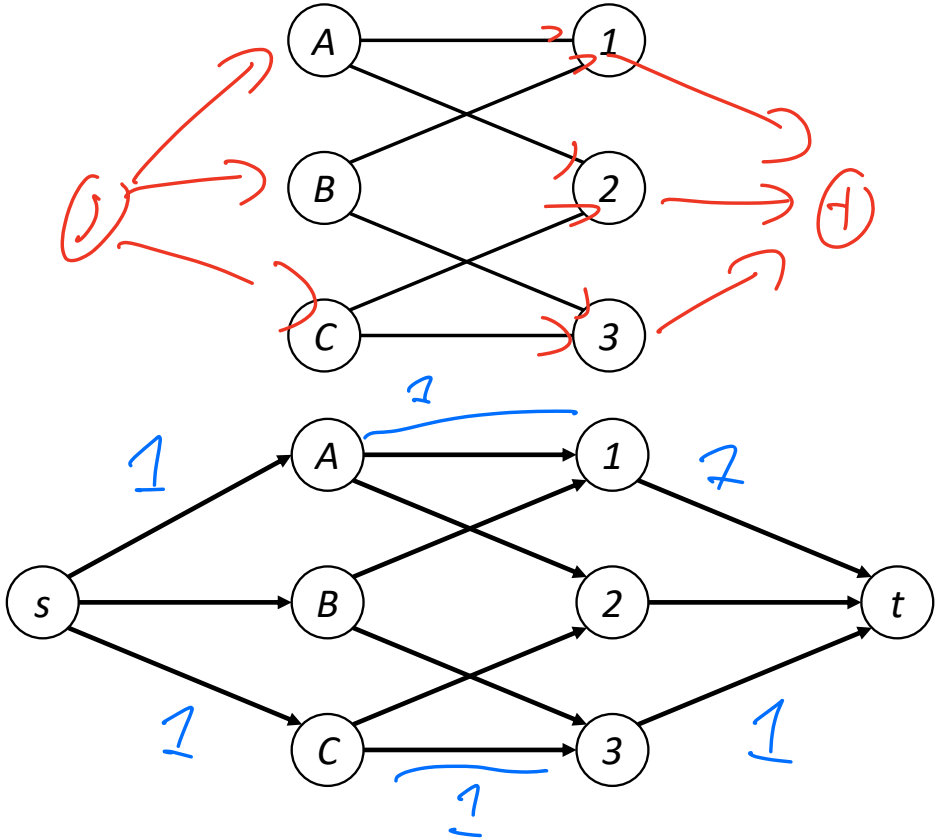
Bipartite Maximum Matching: Given bipartite graph $G = (V, E)$, find matching M maximizing $|M|$

- Extremely important problem, doesn't seem to have much to do with flow!

Algorithm

Give all edges capacity $\mathbf{1}$
Direct all edges from $\boldsymbol{L}$ to $\boldsymbol{R}$
Add source $\boldsymbol{s}$ and sink $\boldsymbol{t}$
Add edges of capacity $\mathbf{1}$ from $\boldsymbol{s}$ to $\boldsymbol{L}$
Add edges of capacity $\mathbf{1}$ from $\boldsymbol{R}$ to $\boldsymbol{t}$

Run FF to get flow $\boldsymbol{f}$
Return $\boldsymbol{M} = \{\boldsymbol{e} \in \boldsymbol{L} \times \boldsymbol{R} : \boldsymbol{f}(\boldsymbol{e}) > \mathbf{0}\}$



Correctness

Claim: M is a matching

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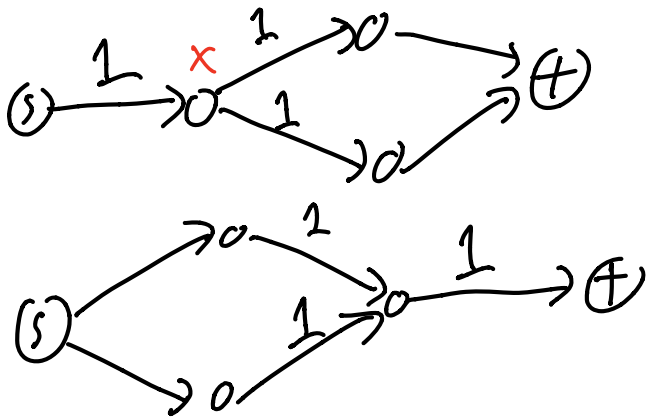
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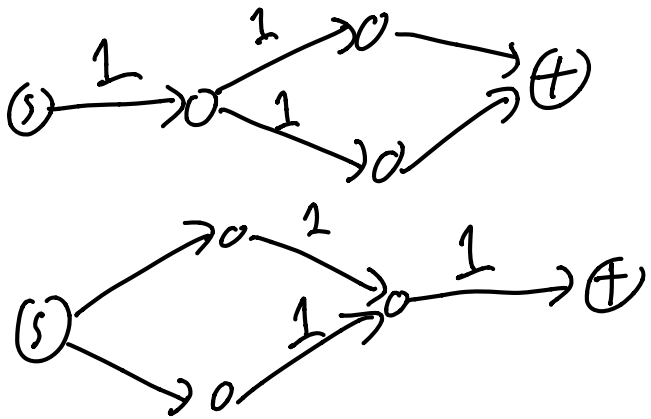
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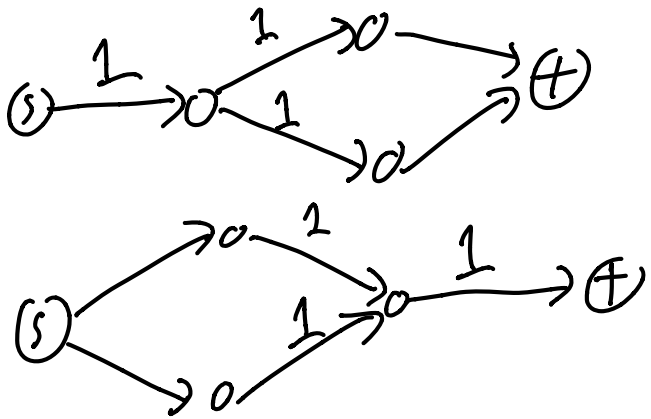


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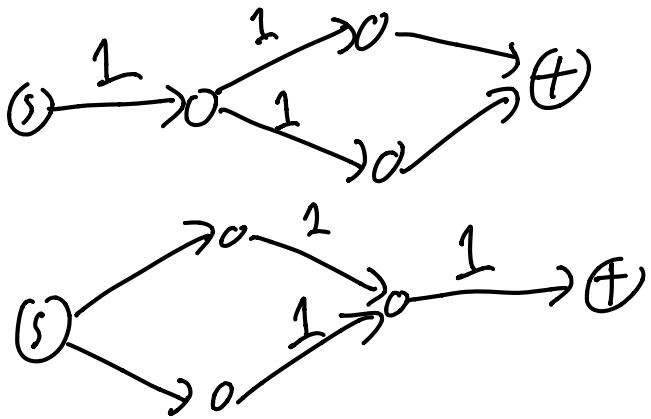
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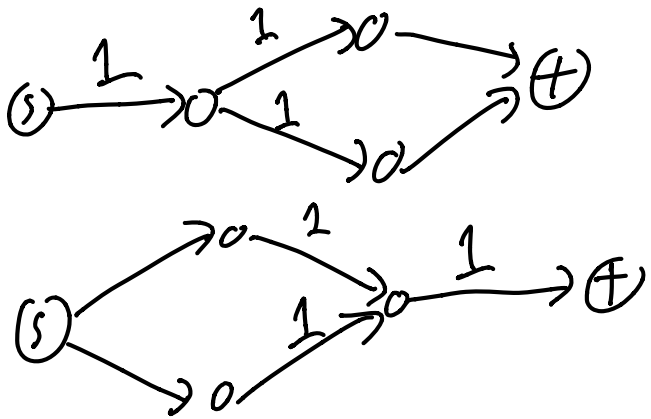
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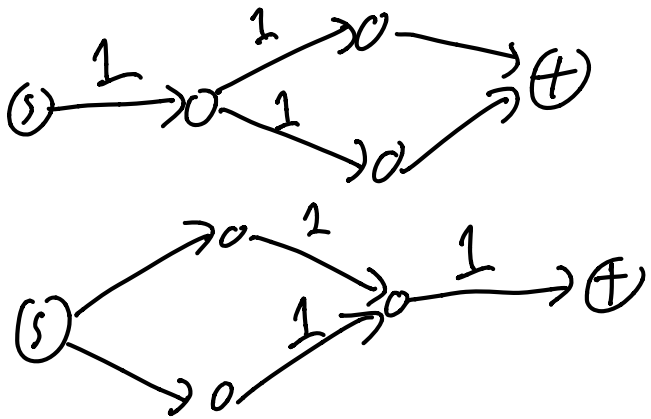
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- $f'(v,t) = 1$ if v matched in M', otherwise 0
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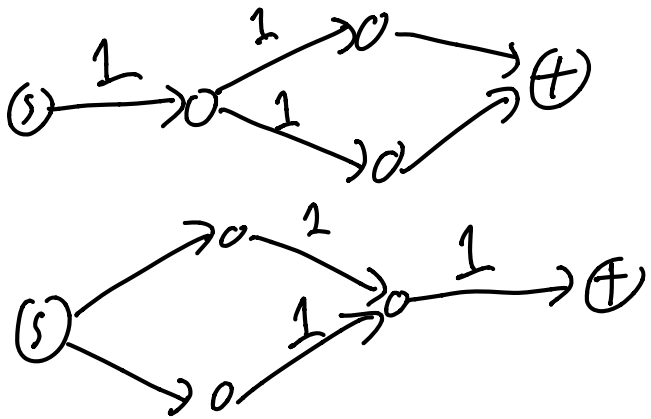
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Running Time

Running Time:

- $O(n+m)$ to make new graph
- $|f| = |M| \leq n/2$ iterations of FF

$\Longrightarrow$ $O(n(m+n)) = O(mn)$ time (assuming $m \geq \Omega(n)$)

Extensions

Many extensions:

- Max-weight bipartite matching
- Min-cost perfect matching
- Matchings in general graphs
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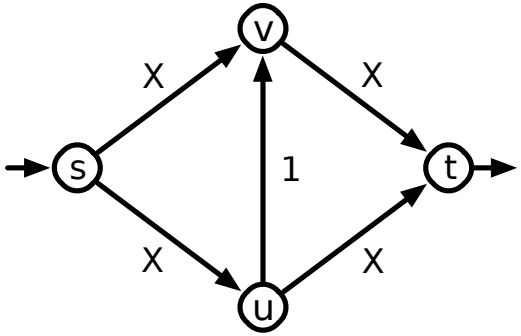
Still active area of study!

- Michael Dinitz, Sungjin Im, Thomas Lavastida, Benjamin Moseley, Sergei Vassilvitskii. *Faster Matchings via Learned Duals*. NeurIPS 2021.
- Michael Dinitz, George Li, Quanquan Liu, Felix Zhou. *Differentially Private Matchings*. Submitted, on arXiv.

Edmonds-Karp

Intuition

Bad example for Ford-Fulkerson:

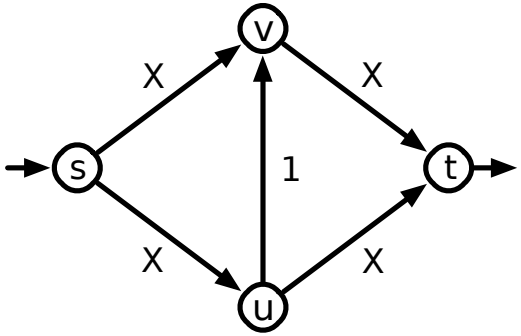


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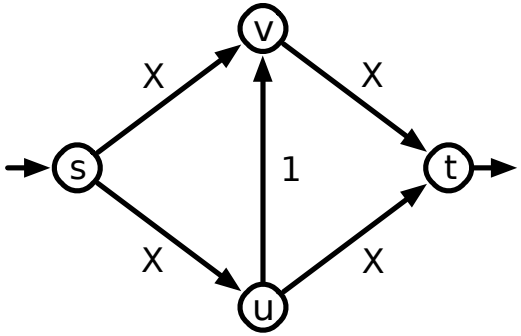
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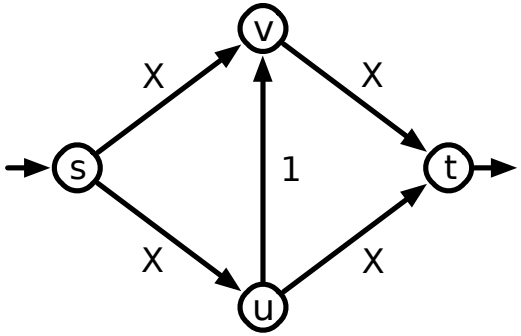
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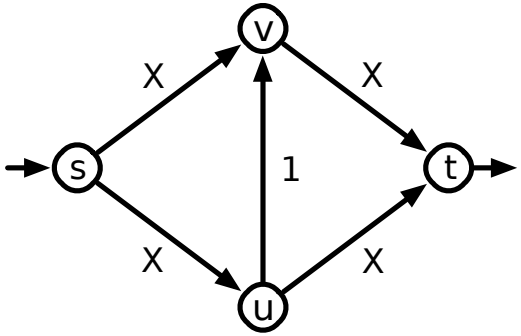
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$$\underset{\text{augmenting paths } P}{\arg\max} \min_{e \in P} c_f(e)$$ ("widest" augmenting path)

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Less obvious path to pick:

$$\underset{\text{augmenting paths } P}{\arg\min} |P| \qquad \text{(augmenting path with fewest edges)}$$

Edmonds-Karp

Use Ford-Fulkerson, but pick *shortest* augmenting path (unweighted)

- Ignore capacities, just find augmenting path with fewest hops!
- Easy to compute with BFS in $O(m+n)$ time.
- Correct, since just FF with particular path choice.

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Theorem

Edmonds-Karp has at most $O(mn)$ iterations, so at most $O(m^2n)$ running time (if $m \geq n$)

Proof (sketch) of Edmonds-Karp

Idea: prove that distance from s to t (unweighted) goes up by at least one every $\leq m$ iterations.

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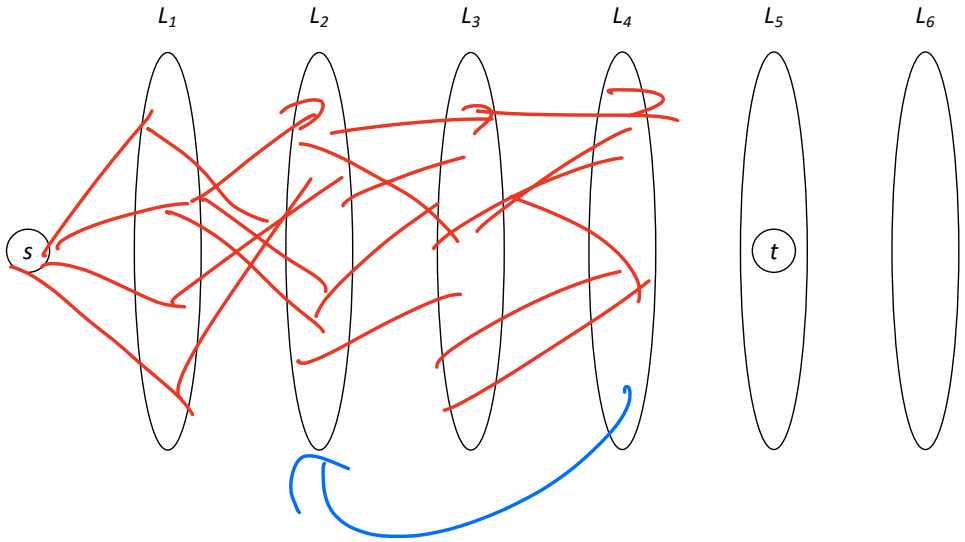
- Distance initially $\geq 1 \implies$ distance $\geq n$ after at most $m(n-1)$ iterations
- Only distance larger than $n-1$ is ∞: no $s \to t$ path

$\implies$ Terminates after at most mn iterations.

Proof (sketch) of Edmonds-Karp (continued)

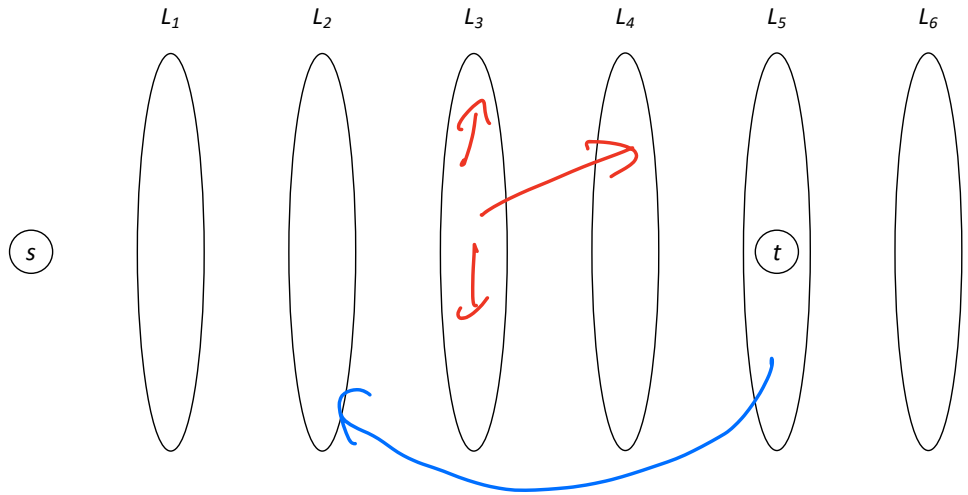
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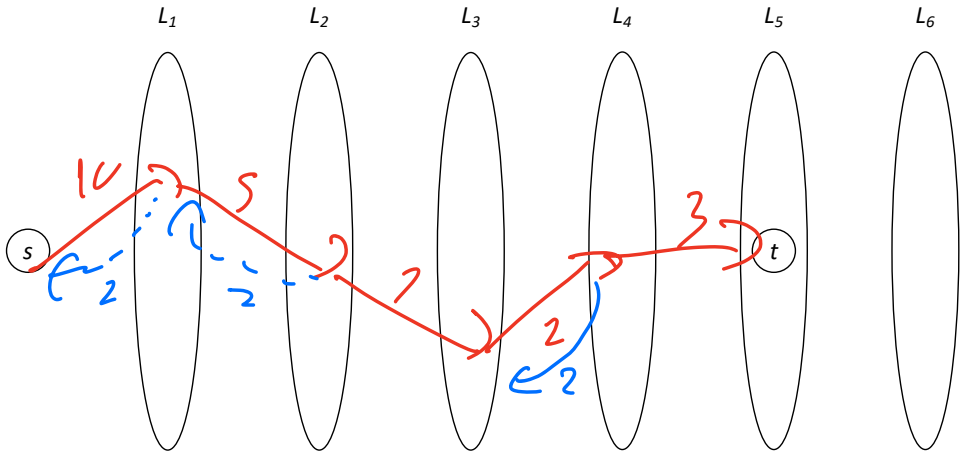


Edge types:

- Forward edges: $\mathbf{1}$ level
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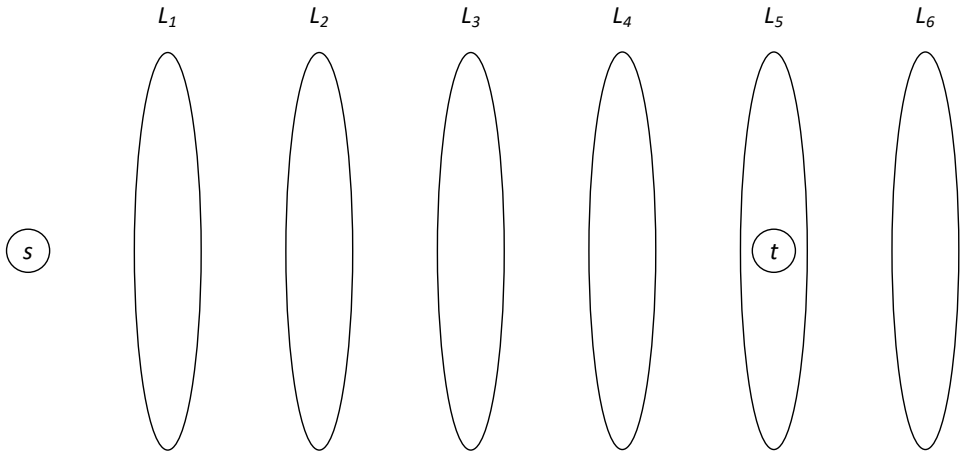
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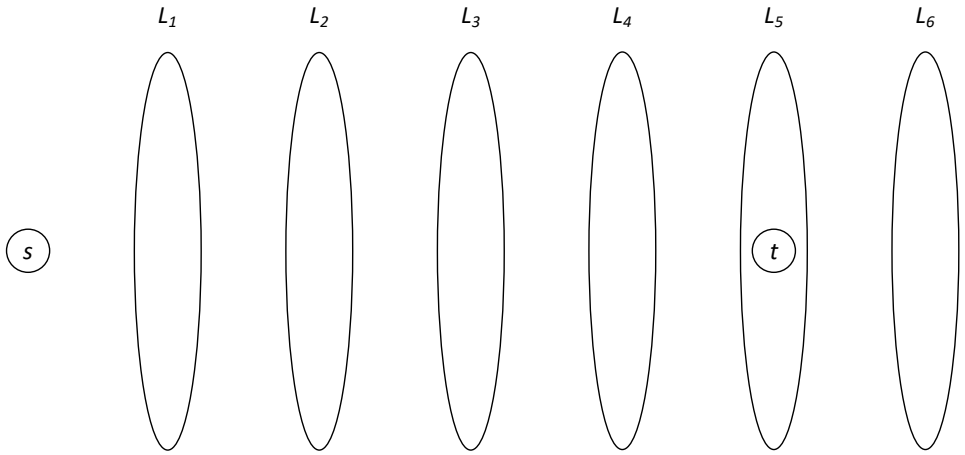
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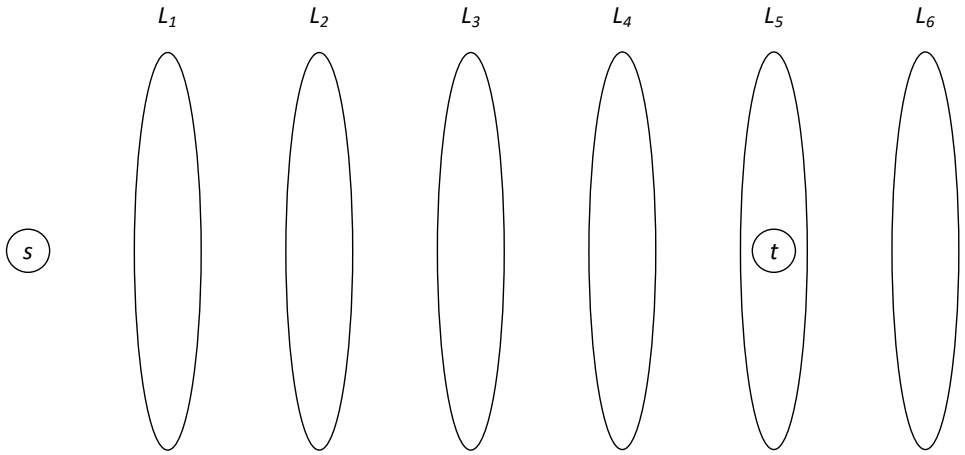
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So after m iterations (same layout): no path using only forward edges $\implies$ distance larger than d!

Finishing Edmonds-Karp

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So at most mn iterations. Each iteration unweighted shortest path: BFS, time $O(m+n)$

Total time: $O(mn(m+n)) = O(m^2n)$. Independent of F!

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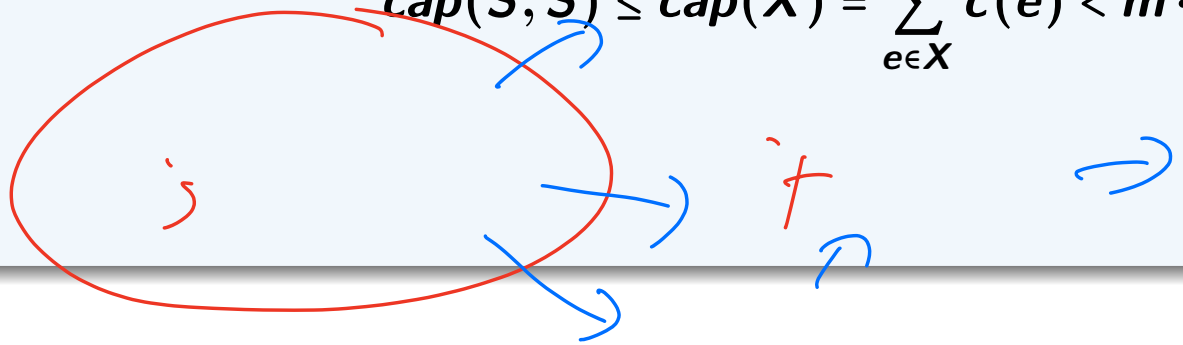
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Does this imply at most m iterations?

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By induction: after iteration i, at most $F(1 - 1/m)^i$ flow remaining to be sent.
Super useful inequality: $1 + x \leq e^x$ for all $x \in \mathbb{R}$

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Theorem

If F is the value of the maximum flow and all capacities are integers, # iterations is at most $O(m \log F)$

How much flow remains to be sent after iteration i?

- $i = 0$: F
- $i = 1$: Sent at least F/m, so at most $F - F/m = F(1 - 1/m)$ remaining
- $i = 2$: Sent at least R/m if R was remaining after iteration 1, so at most $R - R/m = R(1 - 1/m) \leq F(1 - 1/m)^2$ remaining

By induction: after iteration i, at most $F(1 - 1/m)^i$ flow remaining to be sent.
Super useful inequality: $1 + x \leq e^x$ for all $x \in \mathbb{R}$
$\implies$ If $i > m \ln F$, amount remaining to be sent at most

$$F(1 - 1/m)^i < F(1 - 1/m)^{m \ln F} \leq F(e^{-1/m})^{m \ln F} = F \cdot e^{-\ln F} = 1$$

But all capacities integers, so must be finished!

Finishing up

Modified version of Dijkstra: find widest path in $O(m \log n)$ time

- Total time $O(m \log n \cdot m \log F) = O(m^2 \log n \log F)$
- Polynomial time!

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But *weakly* polynomial time: depends on the numbers.

- Edmonds-Karp is *strongly* polynomial time: independent of numbers.

Extensions

Many better algorithms for max-flow: *blocking flows* (Dinitz's algorithm (not me)), *push-relabel* algorithms, etc.

- CLRS has a few of these.
- State of the art:
  - Strongly polynomial: $O(mn)$. Orlin [2013] & King, Rao, Tarjan [1994]
  - Weakly Polynomial: $O(m^{1+o(1)} \log U)$ (where U is maximum capacity). Chen, Kyng, Liu, Peng, Probst Gutenberg, and Sachdeva [2022]
    - Implies $O(m^{1+o(1)})$-time algorithm for max bipartite matching!

Many other variants of flows, some of which are just $s - t$ max flow in disguise!

- Min-Cost Max-Flow: every edge also has a cost. Find minimum cost max-flow.